

Nathan Liu

(925) 786-3685 | nathan.dtlui@gmail.com | linkedin.com/in/n8liu | nliu.org | Danville, CA

EDUCATION

University of California, Berkeley

May 2026

Bachelor of Arts in Computer Science & Data Science

Berkeley, CA

- **Coursework:** Machine Learning, Natural Language Processing, Deep Learning, Data Engineering, Computer Security, Optimization Models in Engineering, Database Systems, Probability for Data Science

TECHNICAL SKILLS

Languages: Python, Java, SQL (PostgreSQL), C/C++, TypeScript, Go

Backend: Django, NumPy, Pandas, Matplotlib, PySpark, Kafka

Cloud/Infra: Kubernetes, AWS (Kinesis, Lambda, S3, DynamoDB, EC2), Docker, Azure, GCP (BigQuery)

Machine Learning: PyTorch, Hugging Face, TensorFlow, LangChain, OpenCV

EXPERIENCE

Data Engineer Intern

May 2025 – Aug. 2025

Carbon Sustain

Berkeley, CA

- Built a real-time LLM decisioning pipeline on AWS (Kinesis, Lambda, Step Functions) to process 100k events/day which enforced schema-validated JSON outputs and wrote full traces to S3 for auditability and replay.
- Improved platform reliability by adding idempotency keys in DynamoDB and CloudWatch alarms to reduce failed executions and eliminated release-related deployment failures by 50%.
- Cut dashboard data retrieval time by 80% by rewriting high-cost joins, adding composite indexes, and implementing cursor pagination in Django/PostgreSQL APIs.

Software Engineer

Jan. 2025 – Aug. 2025

UnifIBD

Remote (San Jose, CA)

- Prototyped a mobile CI/CD pipeline with GitHub Actions, automating deployments to TestFlight + Google Play, which cut manual release times from hours to minutes.
- Deployed a native-feel, in-app chat interface that integrated with Google's Dialogflow API to provide real-time AI support, reducing manual support ticket volume by 30%.
- Built a Django backend and admin CMS that enabled marketing to update in-app content without requiring an app store submission.

Data Science Intern

Jun. 2024 – Aug. 2024

L.A. Lucky Import & Export Inc.

Los Angeles, CA

- Built a near-real-time SAP inventory sync for 3,647+ SKUs using incremental pulls and upserts into Postgres which reduced manual entry by 100% and enabled automated invoicing for all customers.
- Rebuilt a legacy platform (Django REST, JWT, React) and improved performance 75% by reducing payload size and adding server-side caching.
- Built a Docker-based CI/CD workflow to deploy ETL/ELT jobs automatically, streamlining legacy database migration and improving pipeline reliability.

PROJECTS

Clickbait Classifier | *Python, PyTorch, Pandas*

- Fine-tuned a BERT-based classifier on 18K social-news titles; achieved 0.89 F1 and 91% accuracy on a held-out test set, outperforming a TF-IDF logistic regression baseline by +0.14 F1 points.

ItineraryAI | *Python, LangChain, FastAPI, JavaScript*

- Built a chat-based travel concierge that generates structured itineraries from a single prompt and shipped a web dashboard for trip planning (budgets, maps, calendar sync).

The IBD Digest | *Flutter, Dart, Provider*

- Built an offline-first IBD companion app for 5,000+ users (recipes, ingredient libraries, SIBDQ assessment) with persistent onboarding and local state persistence.